

Public Economics and Policy Lecture #5

Theory and Application of Public Finance III (A Two-period Neoclassical Model)

September 1, 2026

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Learning objective

- ❑ **Understand the interaction between consumption, saving, and fiscal policy** --- how the two-period model illustrates the relationship between current and future consumption and saving behavior, and how shifts in fiscal policy influence both current and future economic conditions, including the effects on consumer behavior and savings rates
- ❑ **Analyze equilibrium in the goods market and the impact of fiscal policy on economic equilibrium** --- how changes in fiscal policy, such as increased government spending, impact the supply and demand for goods and services, the labor market, and overall economic equilibrium

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Outline

- ❑ Two-period Model
 - ❑ The Permanent Level of Fiscal Variables
 - ❑ Consumption and Saving Behavior
 - ❑ The Labor Market and Supply Function
 - ❑ Equilibrium in the Goods Market
- ❑ The Effect of Fiscal Policy

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Reference

- ❑ Ihori, T. (2017). *Principles of public finance*. Springer Singapore. (Chapter 3)

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Time factor involved when making decisions across time periods

- ☐ 2 choices: (a) Receive 100 baht today; or,
(b) Receive 100 baht one year later
- ☐ Which one would you choose?

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Time factor involved when making decisions across time periods

- ☐ 2 choices: (a) Receive 100 baht today; or,
(b) Receive 100 baht one year later
- ☐ Which one would you choose?
- ☐ And, having more information ...
 - ☐ Given that the inflation rate was 10%, would you choose (a) or (b)?
 - ☐ Given that the inflation rate was zero, would you choose (a) or (b)?

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M is the PV of 200 baht.

Suppose $r = 5\%$ ($r = 0.05$)

200 baht



At time 0



At time 1

Now

- Present value (PV) is the current value of a future sum of money given a specified rate of return.
- To find the present value of the above example, we follow these steps.

Principal + returns = sum of money at time 1

$$M + (M \times 0.05) = 200$$

$$M(1 + 0.05) = 200$$

$$M = \frac{200}{(1 + 0.05)}$$

The present value of 200 baht is

$$M = \frac{200}{(1 + r)}$$

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M is the PV of 200 baht.

Suppose $r = 5\%$

200 baht



At time 0



At time 2

Now

- To find the present value of the above example, we follow these steps.

Principal + returns = sum of money at time 2

$$M(1 + 0.05)(1 + 0.05) = 200$$

$$M = \frac{200}{(1 + 0.05)^2}$$

The present value of 200 baht is

$$M = \frac{200}{(1 + r)^2}$$

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Review: The Present Value of the Earnings Stream

- ❑ The present value (PV) of a payment of y dollars next year is given by

$$PV = \frac{y}{1+r}$$

- ❑ The PV of receiving y dollars two years from now is

$$PV = \frac{y}{(1+r)^2}$$

- ❑ The PV of receiving y dollars t years from now is

$$PV = \frac{y}{(1+r)^t}$$

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A Quick Look at the Two-period Neo-classical Model

- ❑ A simplified version of the neoclassical growth model that assumes the economy has only two periods: the present and the future.
- ❑ The model analyzes how households and firms make **intertemporal decisions** about consumption, saving, investment, and production, given their preferences, technology, and budget constraints.
- ❑ **Households** supply labor in the labor market. Two choices of households: consumption and labor.
- ❑ **Firms** employ labor with physical capitals in the production of goods.
- ❑ The model can be solved by finding the **optimal choices** of households and firms in each period, subject to their constraints, and then finding the market equilibrium conditions that clear the markets for goods, capital, and labor in each period.

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Example: How to achieve a life-time income of \$95.5

(A) Given that $r = 10\%$
Y is 50 for every period.

(B) Given that $r = 10\%$
Y at the current period is 10.
Y in the future is 94.

PV of the sum of Y for two periods
is

$$50 \times \left(1 + \frac{1}{1+r} \right)$$

PV of the sum of Y for two periods
is

$$10 + \left(\frac{1}{1+r} \times 94 \right)$$

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Example: How to achieve a life-time income of \$95.5

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(B) Given that $r = 10\%$
Y at the current period is 10.
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PV of the sum of Y for two periods is

$$\begin{aligned} & 50 \times \left(1 + \frac{1}{1+r} \right) \\ & 50 \times \left(1 + \frac{1}{1+0.1} \right) \\ & = 50 \times (1.91) \\ & = 95.5 \end{aligned}$$

PV of the sum of Y for two periods is

$$\begin{aligned} & 10 + \left(\frac{1}{1+r} \times 94 \right) \\ & 10 + \left(\frac{1}{1+0.1} \times 94 \right) \\ & = 95.5 \end{aligned}$$

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Example: How to achieve a life-time income of \$95.5

(A) Given that $r = 10\%$
Y is 50 for every period.

PV of the sum of Y for two periods is

$$50 \times \left(1 + \frac{1}{1+r}\right)$$

$$\textcircled{50} \times \left(1 + \frac{1}{1+0.1}\right)$$

$$= 50 \times (1.91)$$

$$= 95.5$$

(B) Given that $r = 10\%$
Y at the current period is 10.
Y in the future is 95.

PV of the sum of Y for two periods is

$$10 + \left(\frac{1}{1+r} \times 94\right)$$

$$\textcircled{10} + \left(\frac{1}{1+0.1} \times \textcircled{94}\right)$$

Now Future

$$= 95.5$$

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Example: How to achieve a life-time income of \$95.5

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Y is 50 for every period.

PV of the sum of Y for two periods is

$$50 \times \left(1 + \frac{1}{1+r}\right)$$

$$\textcircled{50} \times \left(1 + \frac{1}{1+0.1}\right)$$

$$= 50 \times (1.91)$$

$$= 95.5$$

(C) Given that $r = 10\%$
Y at the current period is 80.
Y in the future is 17.

PV of the sum of Y for two periods is

$$80 + \left(\frac{1}{1+r} \times 17\right)$$

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Key Takeaways

- **In the Keynesian world**, the government spends and the economy grows, because people spend whatever is in their pocket this year.

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- **In the Neo-Classical two-period world**, households are forward-looking: they know that every baht the government spends must eventually be paid for by them in taxes, and they plan their consumption around their whole lifetime resources rather than around this year's pay slip.
- That single change of assumption turns the fiscal multiplier from something bigger than one into something between zero and one — and, in one case, into a negative number.
- What matters is no longer how much the government spends, **but whether the spending is temporary or permanent, and whether it happens now or is merely promised for later.**

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1. The Permanent Level of Fiscal Variables

1.1 Definition of the Permanent Level of an Economic Variable

- ☐ In the Neoclassical model, households behave rationally over time. (i.e. they may anticipate future economic conditions, including future fiscal policy, and current fiscal policy.)
- ☐ For simplicity, imagine a **two-period model** with **current (period 1)** and **future (period 2)** periods.
- ☐ The permanent level of an economic variable Y is **Y_p** and is defined as the value that has the same value in both periods.
- ☐ The **present value** is also equal to that of the original values, Y_1 and Y_2 . **r** is the rate of interest.

$$Y_p \left[1 + \frac{1}{1+r} \right] = Y_1 + \frac{1}{1+r} Y_2. \quad (3.1)$$

Source: Ihori, T. (2017). *Principles of public finance*. Springer Singapore. (Chapter 3)

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1. The Permanent Level of Fiscal Variables

1.2 The Government's Budget Constraint

- Permanent levels of govt spending and taxes are G_p and T_p

$$G_p \left[1 + \frac{1}{1+r} \right] = G_1 + \frac{1}{1+r} G_2 \text{ and}$$

$$T_p \left[1 + \frac{1}{1+r} \right] = T_1 + \frac{1}{1+r} T_2$$

where G_1 and T_1 are government spending and taxes for period 1 respectively, and G_2 and T_2 are government spending and taxes for period 2 respectively.

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1. The Permanent Level of Fiscal Variables

- The government's budget constraints in both periods are given as

$$G_1 = T_1 + B \text{ and} \quad (3.2)$$

$$(1+r)B + G_2 = T_2 \quad (3.3)$$

- **A budget deficit occurs in period 1 as an amount of $G_1 - T_1$.**
- Public debt in period 1 is equal to B . *A government bond is not wealth. It is a tax invoice with a delivery date on it*
- **One-period maturity public debt** has to be redeemed in period 2, the amount of $(1+r)B$ is needed for debt redemption.
- Government spending in period 2 is G_2 .
- **In period 2, the total spending is $(1+r)B + G_2$ and it is financed by raising taxes.**
- **Since period 2 is the end period in the two-period model, the government cannot issue new bonds in period 2.**

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1. The Permanent Level of Fiscal Variables

- ❑ Eliminating B from Eqs. (3.2) and (3.3), government budget constraint in the present value is given as

$$G_1 + \frac{1}{1+r}G_2 = T_1 + \frac{1}{1+r}T_2. \quad (3.4)$$

Using the notation of permanent variables, this equation is equivalent to

$$G_p = T_p \quad (3.4')$$

- ❑ **The permanent level of government spending must be equal to the permanent level of tax revenue**, irrespective of the size of bond issuance /the amount of public debt.



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2. Consumption and Saving Behavior

2.1 Optimization Over Time

- ❑ Households' consumption behavior is derived by optimization over time.
- ❑ Using a **simple two-period model**, **households' forward-looking behavior can be explained by**

$$U = U(C_1, C_2) = V(C_1) + \frac{1}{1+\rho}V(C_2) \quad (3.5)$$

where **ρ is the rate of time preference**.

- ❑ When **ρ is high**, the **agent evaluates the utility from present consumption more than the utility from future consumption**. (the agent is **less patient**.)

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2. Consumption and Saving Behavior

❑ The budget constraint in each period is given respectively as

In the 1st period $Y_1 = C_1 + S + T_1$ and (3.6)

In the 2nd period $Y_2 + (1+r)S = C_2 + T_2$ (3.7)

where Y_1 and Y_2 denote labor income in each period. S denotes savings. For simplicity, these incomes are exogenously fixed.

❑ Equation (3.6) means that income in the first period Y_1 may be divided between first period consumption, saving, and tax payment.

❑ Equation (3.7) means that the second period income Y_2 plus interest income and savings $(1+r)s$ may be divided between second period consumption C_2 and tax payment T_2 .

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2. Consumption and Saving Behavior

❑ By eliminating S from these two equations, we have the present value budget constraint of a private agent as

$$C_1 + \frac{1}{1+r}C_2 = Y_1 - T_1 + \frac{1}{1+r}(Y_2 - T_2) = (Y_p - T_p) \left(1 + \frac{1}{1+r}\right). \quad (3.8)$$

The present value of consumption

the present value of disposable income

$$C_p = \cancel{Y_p} - T_p \quad (3.8')$$

❑ The **PV of consumption**, the left-hand side of Eq. (3.8), is equal to the **PV of disposable income**, the right-hand side of Eq. (3.8), which is equal to the **PV of permanent income** minus permanent government spending; namely, **the PV of permanent disposable income**.

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Key Point: The tax that changes your behavior is the one you expect, not the one you have already paid.

- Permanent income is the steady amount per period that is equivalent to the household's actual lifetime earnings. Permanent taxes are the steady amount per period equivalent to its lifetime tax bill. Subtract one from the other and you have permanent disposable income — the household's true, sustainable standard of living.
- Consumption in this model tracks permanent disposable income, not current income. A civil servant who receives a one-off bonus of 50,000 baht raises consumption a little, in every year of the rest of her life. A civil servant who is promoted, permanently, by 50,000 baht a year raises consumption by roughly the full amount, immediately.
- Because the government's budget must balance in present value, any rise in permanent government spending is a rise in permanent taxes. And any rise in permanent taxes is a fall in permanent disposable income — which means a fall in consumption, starting immediately, even if not one baht of extra tax has yet been collected.

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2. Consumption and Saving Behavior

□ Households determine first-period and second-period consumption so as to maximize the utility given by Eq. (3.5) subject to Eq. (3.8).

Objective function of a household $U = U(C_1, C_2) = V(C_1) + \frac{1}{1+\rho} V(C_2) \quad (3.5)$

Subject to $C_1 + \frac{1}{1+r} C_2 = Y_1 - T_1 + \frac{1}{1+r} (Y_2 - T_2) = (Y_p - T_p) \left(1 + \frac{1}{1+r} \right). \quad (3.8)$

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2. Consumption and Saving Behavior

2.2 The Permanent Income Hypothesis

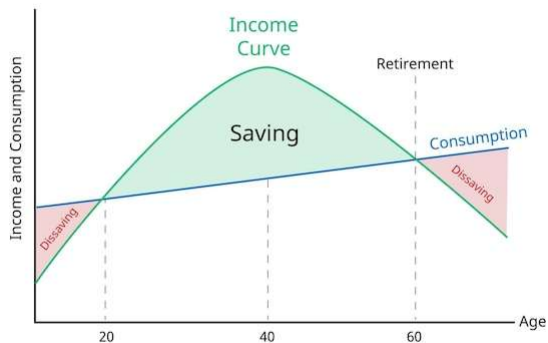
- ❑ Households consume permanent disposable income in each period. → implying that the optimal level of consumption is the same over time. (**Consumption-smoothing behavior**)
- ❑ The time preference (ρ) and the interest rate (r) are not always equalized.
- ❑ If the **interest rate (r) becomes higher**, future consumption is evaluated more than present consumption.
- ❑ If the **time preference rate (ρ) is higher**, present consumption is evaluated more than future consumption
- ❑ In the long run, **consumption may increase as economic growth occurs**. If the rate of interest is greater than the rate of time preference ($r > \rho$), future consumption rises.
- ❑ **Consumption depends only upon permanent disposable income.** (different from the Keynesian consumption function)



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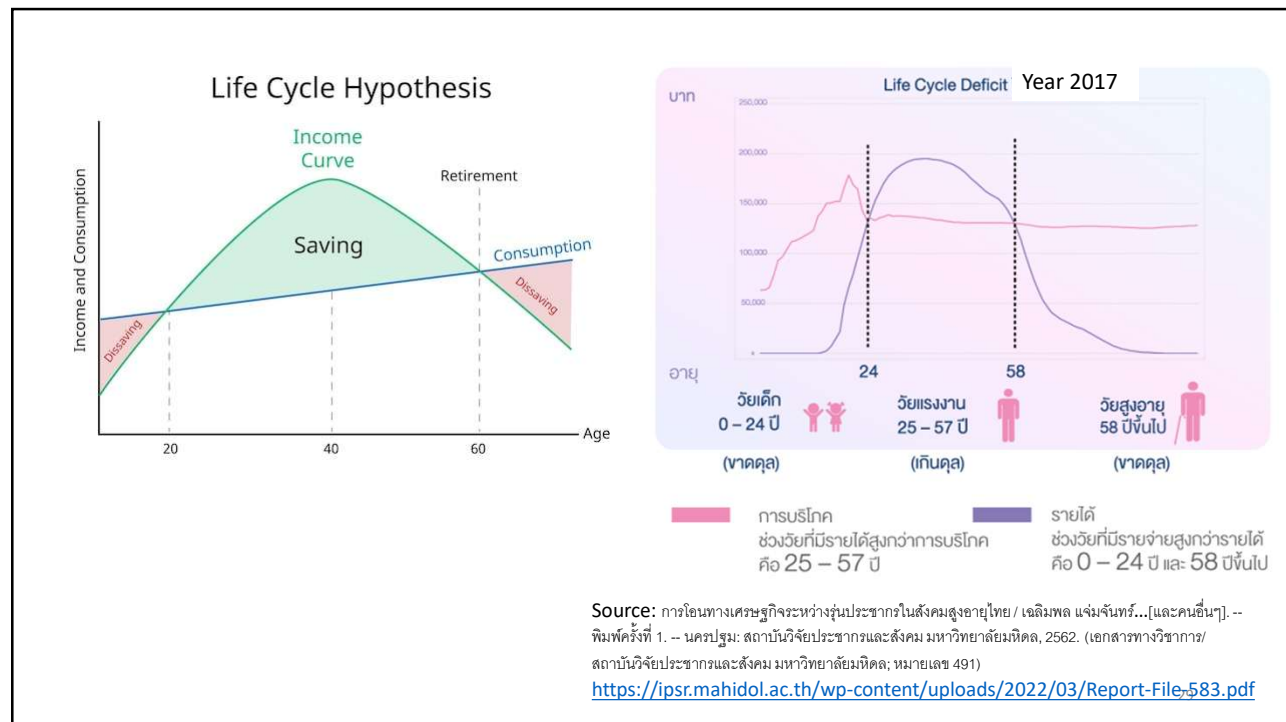
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Life Cycle Hypothesis



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Explanation

- Consumption-smoothing is the idea that people try to keep their standard of living steady over time—even when their income goes up or down.
- Instead of spending a lot when they earn more and cutting back drastically when they earn less, they save during good times and use those savings during tough times. It's like trying to keep your lifestyle on cruise control, even when the road gets bumpy.

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Everyday Examples

1. The Freelancer's Rollercoaster

- Scenario: A freelance designer earns \$5,000 in January, but only \$1,000 in February.
- Behavior: Instead of splurging in January and struggling in February, they save a chunk of January's income to cover February's bills.
- Why it matters: This helps them avoid stress and maintain a stable lifestyle.

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Everyday Examples

2. Saving for Retirement

- Scenario: A person earns a regular salary during their working years but expects no income after retirement.
- Behavior: They save money while working so they can continue spending at a similar level after retiring.
- Why it matters: It prevents a sudden drop in living standards when income stops.

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3. The Labor Market and Supply Function

3.1 Labor Supply by Households

- ❑ Time allocation: households optimally allocate leisure and work (labor supply) among the initial holdings of time (24 hours per day).
- ❑ In the labor market, the wage rate is determined so as to attain full employment.

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3. The Labor Market and Supply Function

The standard utility function is given as $U = U(c, L)$ (3.10)

Subject to the budget constraint $c = wL$ (3.11)

c = wL

Spending on consumption
Labor income

where c is consumption and L is labor (**more hours worked, fewer leisure**)

- ❑ Utility U increases with c and decreases with L .
- ❑ **An increase in wage** → raises the relative attractiveness of labor supply compared with leisure → **the substitution effect**. (Labor supply ↑, Leisure ↓)
- ❑ **An increase in wage** → raises the demand for leisure and reduces labor supply (L ↓) since leisure is a normal good. → **the income effect** (Labor supply ↓, Leisure ↑)
- ❑ Thus, **if the substitution effect is greater than the income effect, an increase in wage stimulates labor supply.**

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3. The Labor Market and Supply Function

The effect of the interest rate on labor supply in an intertemporal framework

- The agent has 2 choices: consumption and labor in two periods, the present and a future period.
- An increase in the interest rate ($r \uparrow$) causes an **intertemporal substitution effect** that attracts future goods and leisure consumption compared with current goods and leisure consumption.
 - $r \uparrow$, **future consumption is increased** because of the **substitution effect**. (greater future consumption, fewer current consumption)
 - Leisure is regarded as a component of consumption. $\rightarrow r \uparrow$ implies that **current leisure consumption declines** and **current labor supply increases**.

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3. The Labor Market and Supply Function

- To conclude, households' labor supply increases with the interest rate, r , and wage rate, w .

$$L_s = L_s^+(r, w)^+ \quad (3.12)$$

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3. The Labor Market and Supply Function

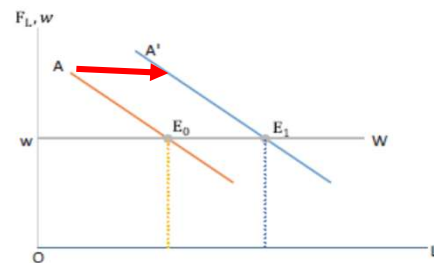
3.2 Labor Demand by Firms

□ When $r \uparrow$, it depresses demand for capital (K).

As capital and labor are substitutable, demand for labor will increase at a given production level.

□ If $K \downarrow$, the marginal product of labor (MP_L) increases. (i.e., it moves the A curve upward to A' , stimulating labor demand.)

Fig. 3.5 Labor demand



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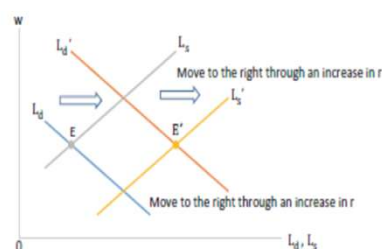
3. The Labor Market and Supply Function

3.3 Equilibrium in the Labor Market (see Figure 3.6)

$$L_s(w, r) = L_d(w, r).$$

When r increases

Fig. 3.6 Equilibrium in the labor market



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3. The Labor Market and Supply Function

- ❑ An increase in r moves the supply curve, L_s to L_s' , to the right. (**Labor supply is an increasing function of r .**)
- ❑ An increase in r also moves the demand curve, L_d to L_d' , to the right. (**Labor demand is an increasing function of r .**)
- ❑ Y_s is an increasing function of r .
- ❑ At the new equilibrium E' , wages do not change and employment (+production) increases.

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4. Equilibrium in the Goods Market

Aggregate supply Aggregate demand

$$Y_s(r) = C(Y_p - G_p) + I(r) + G. \quad (3.18)$$

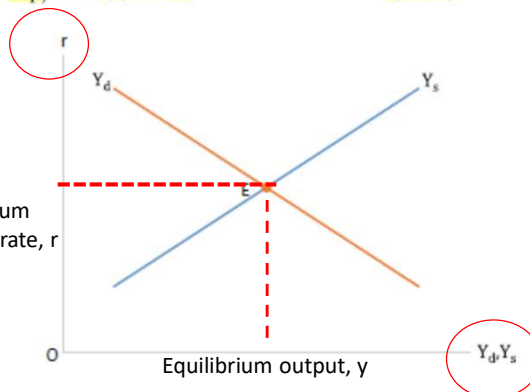
Fig. 3.7 Equilibrium in the goods market

Note:

- ❑ Equilibrium is always at full employment and involuntary unemployment does not exist in the neoclassical model.
- ❑ Consumption depends upon life-time disposable income (not current disposable income) ----- key difference from Keynesian model



Equilibrium interest rate, r



Source: Ihori, T. (2017). *Principles of public finance*. Springer Singapore. (Chapter 3)

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Let's examine the effects of a fiscal expansionary policy
in a two-period model.

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5. The Effect of Fiscal Policy

- ❑ Current spending and/or future spending
- ❑ Current taxes and/or future taxes
- ❑ Household are rational and they optimize consumption and saving plans in consideration of future fiscal changes and current fiscal policy.
- ❑ Three cases of fiscal expansion:
 - (i) **Temporary expansion**: Current $G \uparrow$, permanent G remains unchanged $\rightarrow \Delta G \uparrow, \Delta G_P = 0$
 - (ii) **Permanent expansion**: Current G remains unchanged, permanent $G \uparrow \rightarrow \Delta G \text{ unchanged}, \Delta G_P \uparrow$
 - (iii) **Permanent expansion**: Both $\Delta G \uparrow, \Delta G_P \uparrow$

Source: Ihori, T. (2017). *Principles of public finance*. Springer Singapore. (Chapter 3)

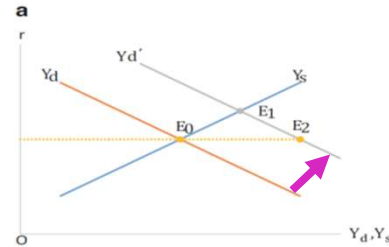
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(i) **Temporary expansion:** Current $G \uparrow$, permanent G remains unchanged $\rightarrow \Delta G \uparrow, \Delta G_P = 0$



- ☐ Future G must fall in order to keep $\Delta G_P = 0$
- ☐ Refer to Figure 3.8a,
- ☐ **Current $G \uparrow$** $\rightarrow$ AD curve shifts to the right $\rightarrow$ the new equilibrium at E_1 ($Y \uparrow$ and $r \uparrow$) (the crowding-out effect of $r \uparrow$ can be seen from E_2 to E_1) (similar to the outcome in the Keynesian model)
- ☐ **Consumption** does not change since permanent income ($Y_P - T_P$) is unchanged.
- ☐ The size of **fiscal multiplier (from E_0 to E_1)** is $0 < \frac{\Delta Y}{\Delta G} < 1$. (Note: Even if r does not change, the size of the multiplier is at most 1 in the Neo-classical model.)



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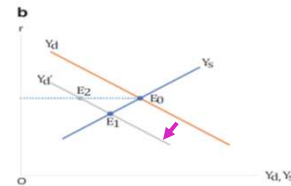
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(ii) **Permanent Expansion:** Current G remains unchanged, permanent $G \uparrow$
 $\rightarrow \Delta G$ unchanged, $\Delta G_P \uparrow$

- ☐ $\Delta G_P \uparrow \rightarrow$ lead to $\Delta T_P \uparrow$ (increase in tax burden) and a decrease in permanent disposable income $\rightarrow \Delta C \downarrow$ (An increase in permanent government spending crowds out private consumption completely.)
- ☐ current $C \downarrow$ and current $Y \downarrow$
- ☐ $\Delta G_P \uparrow$ implies that future $G \uparrow$ and future $Y \uparrow$
- ☐ In all, $\Delta Y_P = 0$ since $\Delta Y_1 \downarrow = \Delta Y_2 \uparrow$
- ☐ Refer to Figure 3.8b,



- ☐ **AD curve** shifts to the left and current Y falls.
- ☐ **$r \downarrow$** $\rightarrow$ crowding-in effect and investment increases
- ☐ The crowding-out effect in consumption outweighs the crowding-in effect in investment $\rightarrow$ negative multiplier
- ☐ **Multiplier** is $-1 < \frac{\Delta Y}{\Delta G_P} < 0$



*** The government would rather raise spending at present than promising to do so in the future. ***

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(iii) **Permanent Expansion**: Both $\Delta G \uparrow$, $\Delta G_P \uparrow$

- $\Delta C \downarrow$ (due to the fall in permanent disposable income) completely offsets $\Delta G \uparrow$
 $\rightarrow$ fiscal multiplier is zero. $\rightarrow \frac{\Delta Y}{\Delta G_P} = 0$ (Aggregate demand does not increase.)
- Note: Government spending crowd outs private consumption and not private investment.



*** From the three cases of fiscal expansion,

The size of fiscal multiplier in the Neo-Classical model is smaller than that in the Keynesian model. ***

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Key Takeaways

- **In the Keynesian world**, the government spends and the economy grows, because people spend whatever is in their pocket this year.

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- **In the Neo-Classical two-period world**, households are forward-looking: they know that every baht the government spends must eventually be paid for by them in taxes, and they plan their consumption around their whole lifetime resources rather than around this year's pay slip.
- That single change of assumption turns the fiscal multiplier from something bigger than one into something between zero and one — and, in one case, into a negative number.
- What matters is no longer how much the government spends, **but whether the spending is temporary or permanent, and whether it happens now or is merely promised for later.**

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Conclusion

- We examined the interaction between current and future consumption, saving, and fiscal policy.
- Households optimize consumption over two periods, balancing between immediate and future consumption based on interest rates and time preference. They aim to maintain stable consumption across periods, influenced by permanent disposable income rather than temporary changes. Changes in interest rates shift both supply and demand for labor, impacting employment and production without affecting wage levels.
- Different fiscal policy changes (temporary vs. permanent increases in government spending) affect aggregate demand, consumption, and savings differently.
- The size of the fiscal multiplier in the neoclassical model is smaller than in the Keynesian model due to the crowding-out effects on private consumption.
- The effectiveness of fiscal policy varies depending on whether changes are temporary or permanent, with significant implications for public finance management.

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